

LECTURE 4

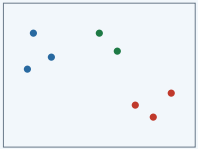
Unsupervised Learning Part 1: Clustering & Dimensionality Reduction

k-Means • DBSCAN • GMM • Hierarchical • PCA • t-SNE • UMAP
Case study: OPFOR tactical archetype discovery, N = 8,000 trajectories, JANUS wargame simulator, 12 features

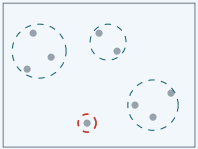
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Supervised: every point labelled



Unsupervised: find the groups

The unsupervised problem

Supervised: "I have labelled examples — learn to distinguish Patrol from Assault" (needs expert annotation of every sample). **Unsupervised:** "Here are 8,000 trajectories — what natural groupings exist? Are there tactics we haven't thought to label?"

Key insight for defence: an analyst labelling data in 2024 cannot create categories for tactics invented in 2026. Clustering discovers NOVEL patterns before analysts know to look for them.

Learning objectives

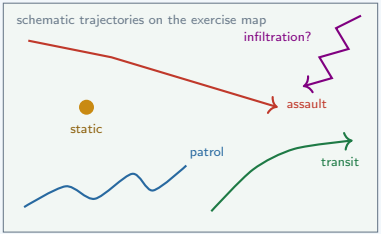
1. Derive k-Means from first principles
2. Apply Elbow / Silhouette for k selection
3. Explain DBSCAN core / border / noise logic
4. Derive GMM via the EM algorithm
5. Compute PCA eigenvectors from the covariance
6. Distinguish t-SNE vs UMAP for visualisation
7. Validate clusters with multiple metrics

Partitional k-Means, GMM	**Density-based** DBSCAN
Hierarchical agglomerative (Ward)	**Dim. reduction** PCA, t-SNE, UMAP

Case Study Dataset — OPFOR Archetype Discovery (JANUS)

Source

JANUS wargame exercise logs (10 armies $\times$ 5-year simulation library). $N = 8{,}000$ force-movement trajectories (800 per army) — **no class labels**: we will discover them by clustering.



Kinematic (8)		Contextual (4)	
x_1	Avg speed (km/h) [0–120]	x_9	Time-of-day code [0–3]
x_2	Max speed (km/h) [0–200]	x_{10}	Weather code [0–4]
x_3	Direction-change rate (deg/min)	x_{11}	Mission-phase code [0–5]
x_4	Formation spread (m)	x_{12}	Force-size category [1–4]
x_5	Dwell time at waypoints (min)		
x_6	Terrain exposure index [0–1]		
x_7	Route deviation (m)		
x_8	Engagement frequency (/h)		

All features are **z-score normalised**: distance-based clustering is meaningless when speed (0–200) and exposure (0–1) are on different scales.

k-Means Clustering — Minimise Within-Cluster Variance (Lloyd's Algorithm)



0 initialise $k = 3$ centres | **1** assign to nearest centre | **2** move centre to cluster mean | **3** repeat until stable

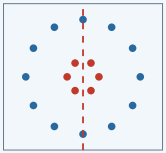
WCSS OBJECTIVE AND THE TWO STEPS

$$J = \sum_{j=1}^{k} \sum_{x_i \in C_j} \|x_i - \mu_j\|^2$$

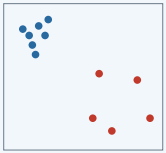
$$c_i = \arg\min_j \|x_i - \mu_j\|^2, \qquad \mu_j = \frac{1}{|C_j|} \sum_{x_i \in C_j} x_i$$

- **Initialise** with k-means++: choose centres with probability $\propto$ squared distance from existing centres
- **Convergence:** J decreases monotonically — guaranteed to converge, NOT to the global optimum (local minima exist)
- **Remedy:** run 10+ times, keep the lowest J

k-Means — Key Properties, Assumptions and Cost



rings: k-Means cuts in half



different densities

What k-Means assumes

✓ convex, roughly spherical clusters ✓ roughly equal size ✓ features on the same scale (z-score!) ✓ Euclidean distance meaningful.

What it cannot do

× arbitrary shapes (rings) × very different densities × determine k automatically × handle noise/outliers.

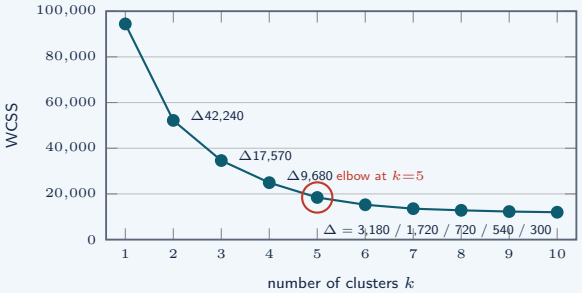
COMPLEXITY

$$O(N \cdot k \cdot d) \text{ per iteration}$$

- Case study: $8000 \times 5 \times 12 = 480$K ops/iteration; 50–150 iterations $\Rightarrow \sim$50M ops, < 0.5 **s** on a laptop
- Random initialisation may converge to a poor local minimum; k-means++ (D^2 seeding) gives $\sim$**16**$\times$ better WCSS in practice
- Always use multiple restarts (`n_init` ≥ 10)

k-Means is the fast first pass — its output is also the standard initialisation for GMM later in this lecture.

Choosing k — the Elbow Method (WCSS vs k)

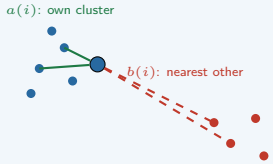


- WCSS **always** falls as k grows: $k = N \Rightarrow \text{WCSS} = 0$; $k = 1 \Rightarrow \text{WCSS} = SS_{\text{total}}$
- Look for the "elbow" — the point of diminishing returns
- Large drops from $k = 2$ to 5 (Δ 42,240 $\rightarrow$ 6,490), small after

Data pitfall: never choose k by minimising WCSS alone. The elbow can be faint or absent — confirm with the silhouette and domain knowledge.

Analogy: adding battalions to a front always shortens each one's sector; the question is when the extra unit stops buying much.

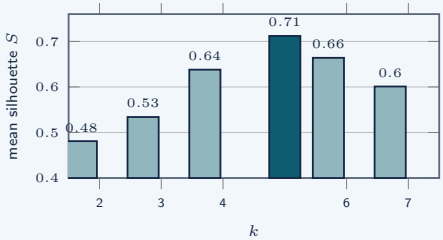
Choosing k — Silhouette Analysis Confirms $k = 5$



SILHOUETTE

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \in [-1, +1], \qquad S = \frac{1}{N} \sum_i s(i)$$

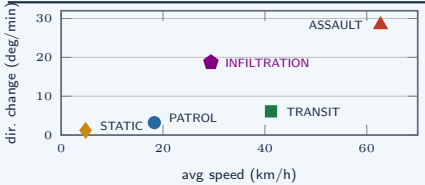
- $s = +1$ well placed; $s = 0$ on a boundary; $s = -1$ likely in the wrong cluster
- $S > 0.70$ strong; $S > 0.50$ reasonable; $S < 0.25$ weak/overlapping



Both methods agree: $k = 5$. Maximum silhouette $S = 0.712$ — strong evidence for five natural archetypes.

$k = 5$ Solution — Five OPFOR Manoeuvre Archetypes Discovered

Cluster	Size	Avg spd	Max spd	Dir chg	Spread	Dwell	Archetype
C-0	1,542	18.3	42.1	3.2°	420 m	22.4 m	PATROL (slow, spread, dwell)
C-1	932	62.7	118.4	28.4°	85 m	1.8 m	ASSAULT (fast, tight, no dwell)
C-2	714	41.2	84.6	6.1°	310 m	4.2 m	TRANSIT (medium, purposeful)
C-3	2,184	4.8	12.3	1.2°	380 m	61.2 m	STATIC / DEFENSIVE (long dwell)
C-4	628	29.4	68.1	18.7°	140 m	8.6 m	**INFILTRATION** (tight, evasive)

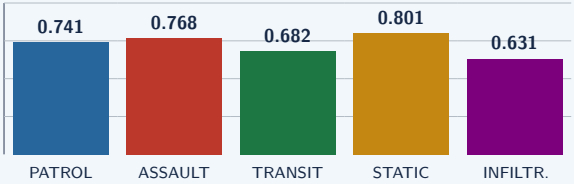


Tactical gold: cluster C-4 (INFILTRATION) was NOT in the original analyst label schema. Moderate speed, tight formation, evasive direction changes, night-biased (time-of-day = 3) — labelled post hoc after domain experts reviewed its members. **Unsupervised ML discovered a new enemy tactic.**

Cluster centroids in two of the 12 dimensions.

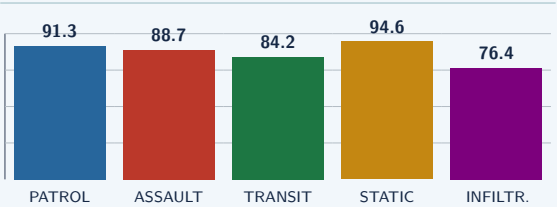
$k = 5$ Validation — Per-Cluster Silhouette and Expert Agreement

Silhouette per cluster (overall mean 0.712)



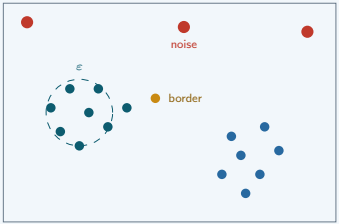
- STATIC is the most distinct (0.801); TRANSIT overlaps with C-4; INFILTRATION overlaps C-1 & C-2 (0.631)

Agreement with 200 analyst-labelled trajectories (%)



Overall **87.0%** expert–cluster alignment. INFILTRATION is lowest — expected, since it is a *new* category the analysts had no prior label for.

DBSCAN — Density-Based Clustering: Any Shape, and Outliers Flagged as Noise



DEFINITIONS

$$\text{core}(p) \iff \left|\{x : d(p,x) \le \varepsilon\}\right| \ge \text{MinPts}$$

cluster = maximal density-connected set

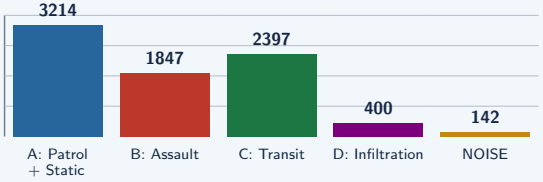
- **Core:** $\ge$ MinPts points within its ε-ball
- **Border:** within ε of a core point but fewer than MinPts neighbours
- **Noise:** neither — labelled as outlier
- **Directly density-reachable:** $d(x,y) \le \varepsilon$ and y is core; **density-connected:** both reachable from a common core point
- **Algorithm:** for each unvisited p: if core, expand the cluster; else mark as noise (may be reassigned)
- **Complexity:** $O(N \log N)$ with a spatial index

Unlike k-Means, DBSCAN discovers k automatically and never forces an odd point into a cluster.

DBSCAN vs k-Means — and the 142 Deception Manoeuvres

	k-Means	DBSCAN
Shape	spherical / convex	**any shape**
Noise	forced into a cluster	**labelled explicitly**
k	required input	**discovered**

Case study ($\varepsilon = 0.45$, MinPts $= 8$)



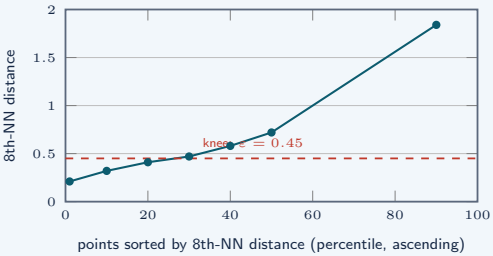
The 142 noise points (1.78%) = deception manoeuvres! Zero dwell, random direction changes, night operations, unusual speed patterns — deliberate adversary deception.

Why this matters

In defence, "noise" is often the most interesting signal: rare or unusual behaviour. k-Means would have forced each of these 142 trajectories into its nearest archetype, hiding them in plain sight.

Note: DBSCAN merged PATROL and STATIC into one dense cluster A — density-based and centroid-based methods see different structure. Use both.

DBSCAN — Tuning ε and MinPts with the k-Distance Plot

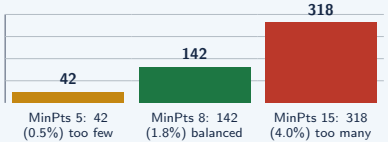


Recast from the reported percentile table (99th%ile of closeness = 0.21 … 10th = 1.84): dense core first, isolated points last.

How to choose ε

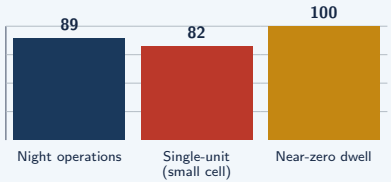
(1) For each point compute the distance to its MinPts-th nearest neighbour; (2) sort ascending; (3) plot; (4) choose ε at the knee (sharp change).

MinPts: noise fraction is sensitive



Rule of thumb MinPts $\geq d + 1 = 13$ for $d = 12$ — but the operational noise rate (1.8%) favoured MinPts $= 8$.

Profile of the 142 noise trajectories

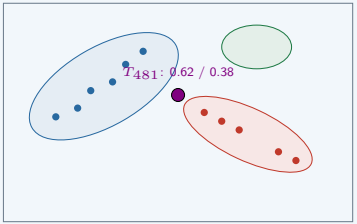


"Near-zero dwell at ALL waypoints" is reported qualitatively; plotted as 100% for illustration.

- Avg direction change **34.2 deg/min** (vs 28.4 ASSAULT, 18.7 INFILTRATION)
- Route deviation **820 m+** from the shortest path

Expert conclusion: DELIBERATE DECEPTION MANOEUVRES — small units moving erratically to confuse pattern-of-life analysis. **Action:** a new OPFOR doctrine SOP for deceptive manoeuvres, added as explicit Category 6 in training.

DBSCAN limitation: struggles with clusters of VARYING density — common in real data (dense urban trajectories vs sparse rural). **HDBSCAN** converts DBSCAN to a hierarchy via a single-linkage tree and extracts stable clusters at multiple density scales: recommended for real intelligence data.



MIXTURE MODEL AND RESPONSIBILITY

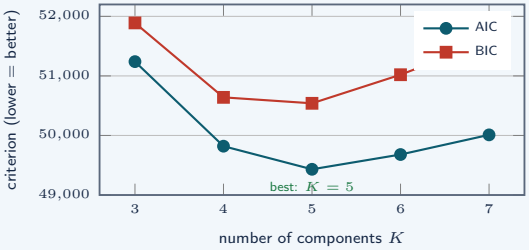
$$P(x) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}(x; \mu_k, \Sigma_k), \quad \sum_k \pi_k = 1$$

$$r_{ik} = \frac{\pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(x_i; \mu_j, \Sigma_j)} \in [0, 1]$$

- π_k: mixture weight; μ_k: mean; Σ_k: covariance (shape and orientation)
- Each point belongs *partially* to ALL clusters: $\sum_k r_{ik} = 1$

	k-Means	GMM
Assignment	hard (one cluster)	**soft (probabilities)**
Shape	spherical ($\Sigma = \sigma^2 I$)	**elliptical (full Σ_k)**
Relation	special case of GMM: $\Sigma_k = \sigma^2 I$, $\sigma \to 0$	

Benefit: trajectory T_{481} has $r(\text{ASSAULT}) = 0.62$, $r(\text{INFILTRATION}) = 0.38 \Rightarrow$ alert analysts that it is *ambiguous* instead of silently labelling it ASSAULT.

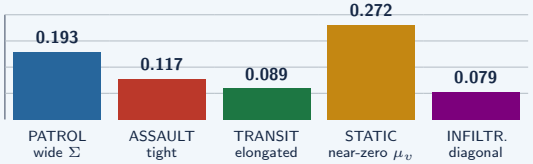


INFORMATION CRITERIA (p PARAMETERS)

$$\text{AIC} = 2p - 2\log L, \qquad \text{BIC} = p\log N - 2\log L$$

BIC penalises extra parameters more heavily than AIC. Here both are minimised at $K = 5$ (AIC 49,432; BIC 50,540; log-likelihood $-24{,}516$). Log-likelihood keeps rising with K ($-24{,}205$ at $K = 7$) — never select on likelihood alone.

$K = 5$ cluster profiles (mixture weights π)



The EM Algorithm — Maximum Likelihood for GMM by Alternating E- and M-Steps

GOAL AND E-STEP

$$\mathcal{L}(\theta) = \sum_{i=1}^{N} \log \sum_{k} \pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k)$$

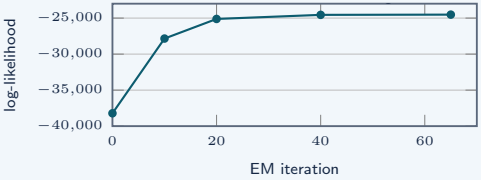
$$\textbf{E:}\quad r_{ik} = \frac{\pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(x_i; \mu_j, \Sigma_j)}$$

M-STEP

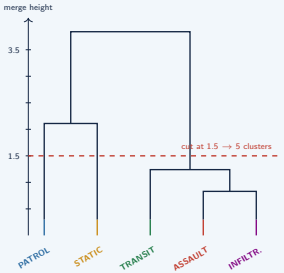
$$N_k = \sum_i r_{ik}, \quad \pi_k = \frac{N_k}{N}, \quad \mu_k = \frac{1}{N_k} \sum_i r_{ik} x_i$$

$$\Sigma_k = \frac{1}{N_k} \sum_i r_{ik} (x_i - \mu_k)(x_i - \mu_k)^\top$$

Initialise θ^0 from k-Means (fast). $\mathcal{L}$ never decreases (EM theorem); stop when $|\Delta \mathcal{L}| < 10^{-6}$.



Pitfall 1 — local maxima: use 10 random starts (best $-24{,}516$). **Pitfall 2 — singularities:** a component collapsing onto one point gives $\Sigma_k \to 0$, likelihood $\to \infty$; regularise $\Sigma_k \mathrel{+}= \varepsilon I$ or bound the minimum eigenvalue ≥ 0.01. **Pitfall 3 — label switching:** cluster "1" differs between runs; align runs with the Hungarian algorithm.



Schematic dendrogram from the reported merge heights (ASSAULT–INFILTRATION 0.83; final merge 3.84).

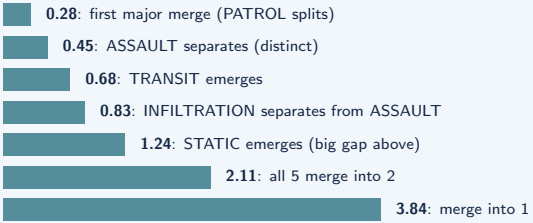
Agglomerative algorithm

Start with N singleton clusters; repeatedly merge the two **closest** clusters until one remains $\Rightarrow$ a binary merge tree (dendrogram).

Linkage	Effect
Single (min pair)	chaining; elongated clusters
Complete (max pair)	compact, spherical
Average (mean pair)	balanced; most common
Ward (min ΔWCSS)	compact; best for defence use

- Naive $O(N^3)$; efficient $O(N^2)$: $N = 8{,}000 \Rightarrow \sim$64M ops $\approx$ 8 s
- Cut at the **largest vertical gap** in the dendrogram

Key merge heights (2,000-trajectory subsample)

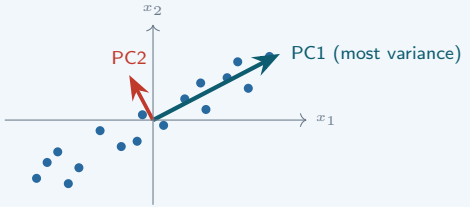


Cut at height 1.5 → 5 clusters — matches k-Means. Adjusted Rand Index between Ward ($k = 5$) and k-Means ($k = 5$): **ARI = 0.847** (1 = identical, 0 = random). Two completely different methods agree ⇒ strong evidence for five real archetypes.

Defence value: a visual intelligence product

The dendrogram shows which clusters are most alike: ASSAULT and INFILTRATION are behaviourally closest (shared speed/formation, merge height 0.83), while STATIC and PATROL are most distinct from all others. This informs COA analysis and watch-list prioritisation.

Principal Component Analysis — Directions of Maximum Variance



PCA rotates the axes to line up with the data's spread, then keeps the first k directions. Linear and unsupervised: it ignores clusters and labels.

DERIVATION

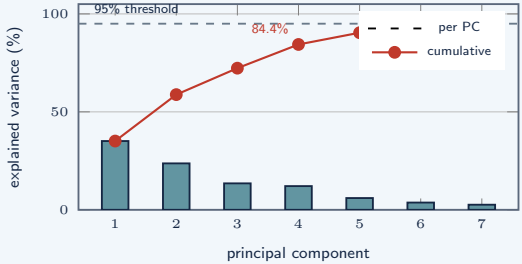
$$\bar{X} = X - \text{mean}(X), \qquad C = \tfrac{1}{N}\bar{X}^\top \bar{X} \in \mathbb{R}^{d\times d}$$

$$C = V\Lambda V^\top, \quad \Lambda = \text{diag}(\lambda_1 \geq \cdots \geq \lambda_d)$$

$$\text{EVR}_j = \frac{\lambda_j}{\sum_i \lambda_i}, \qquad Z = \bar{X}V_k$$

- C is symmetric PSD $\Rightarrow$ real, non-negative eigenvalues; λ_j = variance along PC j
- V_k = top-k eigenvectors; Z = PC scores ($N \times k$)
- **SVD** for numerical stability: $\bar{X} = USV^\top$, $\lambda_j = s_j^2/(N-1)$

PCA on the 12-Feature Case Study — Four Components Explain 84.4%



Eigenvalues: 4.21, 2.84, 1.62, 1.45, 0.72, 0.44, 0.31, . . . , 0.03 (PC12).

Kaiser rule ($\lambda > 1$): keep 4 PCs (84.4%). **95% threshold:** 6 PCs.

PC1 = "Activity / Aggression" axis

Loadings: speed 0.48, max speed 0.41, direction change 0.38.

PC2 = "Dispersion / Caution" axis

Loadings: spread 0.52, dwell time 0.44, exposure 0.31.

Naming components from their loadings turns 12 raw features into two *interpretable* tactical dimensions for briefings.

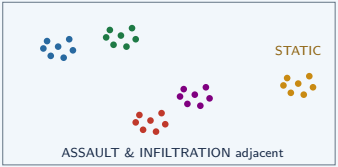
T-SNE: MATCH NEIGHBOURHOODS

$$p_{j|i} \propto \exp\big(-\|x_i - x_j\|^2 / 2\sigma_i^2\big), \quad p_{ij} = \frac{p_{j|i} + p_{i|j}}{2N}$$

$$q_{ij} \propto (1 + \|y_i - y_j\|^2)^{-1}, \qquad C = \sum_{i,j} p_{ij} \log \frac{p_{ij}}{q_{ij}}$$

- PCA is linear — cannot unfold rings or manifolds
- Student-t (heavy tails) in 2-D; minimise KL by gradient descent ($\sim$1000 iterations); **perplexity** 5–50 (typ. 30)
- Case study (2,000 trajectories): 5 distinct clouds; INFILTRATION splits slightly from ASSAULT; STATIC isolated

Warning: distances BETWEEN clusters, and cluster sizes/shapes, in a t-SNE plot are NOT meaningful. Only local structure is preserved.



	t-SNE	UMAP
Runtime (N=8,000)	38.4 s	**4.2 s** (9×)
Global structure	weak	**partly kept**
Deterministic	no	**yes**
Project new points	no	**yes**

UMAP ($n_neighbors = 15$, $min_dist = 0.1$): ASSAULT and INFILTRATION adjacent, STATIC maximally separated — both correct. It can project NEW trajectories into the existing map in real time.

Cluster Validation — How Do We KNOW the Clusters Are Real?

Metric	Formula / method	Range	Value (k=5)	Interpretation
Silhouette S	$(b-a)/\max(a,b)$, mean	-1 to $+1$	0.712	strong (> 0.70)
Calinski–Harabasz	between / within-cluster variance	0 to ∞	1,842	higher = better
Davies–Bouldin	avg max ratio within/between	0 to ∞	0.68	lower = better
Dunn index	min inter / max intra distance	0 to ∞	0.41	higher = better
Adjusted Rand	agreement with expert labels	-1 to $+1$	0.870	near-perfect alignment
Bootstrap stability	re-cluster 100 samples, avg ARI	0 to 1	0.893	very stable

Gap statistic

$\text{Gap}(k) = E[\log \text{WCSS}_{\text{random}}(k)] - \log \text{WCSS}_{\text{data}}(k)$: compares clustering against a random baseline; choose the k with maximum gap. Case study: peaks at $k = 5$.

Prediction strength (Tibshirani 2005)

Split data, cluster both halves, measure how many co-assigned training pairs stay together in the test half; choose k with PS > 0.80. Case study: **PS = 0.887** at $k = 5$.

Verdict: all validation measures agree — the $k = 5$ clustering is robust, stable and expert-aligned (silhouette 0.712, ARI 0.870, bootstrap stability 0.893): a high-confidence solution for operational OPFOR characterisation.

Algorithm Comparison — Four Clustering Methods on 8,000 OPFOR Trajectories

Algorithm	k found	Silhouette	ARI	Noise	Train	Key strength
k-Means (k=5)	5 (input)	**0.712**	**0.870**	none	0.8 s	fast, interpretable centroids
DBSCAN (ε=0.45, m=8)	4 (auto)	0.684	0.821	142 (1.8%)	2.1 s	finds noise; any shape
GMM (K=5)	5 (input)	0.698	0.857	none	18.4 s	soft memberships; ellipses
Ward hierarchical	5 (cut)	0.701	0.847	none	8.3 s	dendrogram for analysts

1. k-Means fast first pass → **2. GMM** initialised from k-Means: probabilities → **3. Hierarchical** visual confirmation → **4. DBSCAN** noise = deception

Shapes handled

k-Means spherical; GMM elliptical; **DBSCAN any**; Ward ≈ spherical.

Finds k automatically?

k-Means no; GMM only via AIC/BIC; **DBSCAN yes**; hierarchical yes (cut).

Scales to large N?

k-Means $O(Nkd)$; GMM $O(NK^2d^2)$; DBSCAN $O(N \log N)$; hierarchical **no**: $O(N^2)$ memory.

Finding structure without labels is one of the oldest ideas in science

BIOLOGY
Taxonomy and phylogenetic trees are hierarchical clusterings of species; gene-expression clustering with heat-map dendrograms (Eisen et al., 1998) found co-regulated genes.
Defence: order-of-battle hierarchies inferred from observed co-movement.

ENGINEERING
k-Means is Lloyd's algorithm, devised at Bell Labs (1957) to quantise signals for pulse-code modulation — vector quantisation and codebooks.
Defence: compressing ISR imagery over narrow UAV links; emitter codebooks in SIGINT.

PSYCHOLOGY
Spearman's general factor g (1904) and the "Big Five" personality traits reduce many test scores to a few latent dimensions — the ancestor of PCA-style reduction.
Defence: summarising dozens of readiness indicators into a few axes.

AERONAUTICS
Airline flight-data monitoring clusters flight-recorder traces to find approaches unlike the rest — before they become accidents.
Defence: sortie debriefs and fleet safety; unusual profiles flagged automatically, like DBSCAN noise.

STATISTICS / EPIDEMIOLOGY
John Snow's 1854 London cholera map: deaths clustered around the Broad Street pump — spatial clustering that located a cause.
Defence: incident-hotspot analysis (e.g. IED events) with density-based clustering.

WARGAMING
Clustering thousands of simulated runs reveals the few distinct ways a battle unfolds — far more useful to a commander than one "average" outcome.
Use: scenario archetypes for course-of-action comparison.

Common thread: clusters are hypotheses, not facts. Snow still had to remove the pump handle; analysts still had to confirm the INFILTRATION archetype.

Lecture 4 Summary — Four Clustering Methods and Three Dimensionality-Reduction Techniques

k-MEANS
Minimise WCSS by alternating assign + update; choose k via elbow + silhouette; fast and interpretable.
Use: first pass; best starting point

DBSCAN
Core / border / noise via ε-neighbourhoods; any shape; 142 noise points = deception manoeuvres.
Use: anomaly detection; no k needed

GMM
Soft memberships r_{ik} fitted by EM; elliptical clusters (full Σ); AIC/BIC for K.
Use: ambiguous boundaries

HIERARCHICAL (WARD)
Dendrogram = visual intelligence product; ARI 0.847 with k-Means; shows cluster relationships.
Use: analyst visualisation

PCA
Linear reduction via covariance eigenvectors; 4 PCs explain 84.4%; PC1 activity, PC2 dispersion.
Use: compression; noise removal

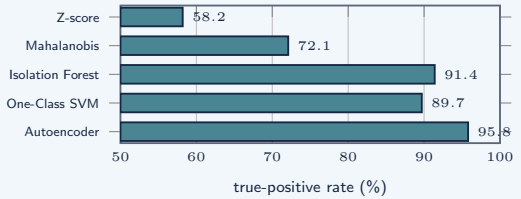
t-SNE & UMAP
Non-linear 2-D maps of clusters; UMAP 9× faster and projects new points.
Caution: t-SNE distances are not meaningful

When "normal" is the label — detecting the abnormal

- Anomaly detection vs clustering
- Statistical methods: z-score, IQR, Mahalanobis distance; robust PCA
- **Isolation Forest:** random recursive partitioning; score = average path length; $O(N)$ training
- **One-Class SVM:** tight RBF boundary around normal; ν controls the outlier fraction
- **Autoencoders:** encode → bottleneck → decode; high reconstruction error = anomaly
- **Cyber:** SIGINT intrusion, lateral movement, command-and-control traffic

Assignment 4: k-Means derivation + DBSCAN noise analysis + one GMM EM step worked by hand.

Preview: network intrusion, $N = 500{,}000$ packets (3.6% attacks, 41 features)



False-positive rates: 0.6% (AE), 0.9% (OCSVM), 1.1% (IF), 1.8% (Mahalanobis), 2.4% (z-score).

Insider threat module

15 compromised accounts among 99.997% normal activity: the autoencoder detects **13/15 (86.7%)** within 48 h of the first anomalous login pattern.